

Quotient Laws of Exponents

MA.7.NSO.1.1

I can use the quotient of powers, power of a quotient and zero exponent laws to rewrite and evaluate a numerical expression.

Yesterday: same base, add the exponents.

Today: same base, subtract — and exponent 0 is not 0.

Warm-Up

On your own. Four minutes. No notes.

1. Write as one power: $3^2 \cdot 3^3$
2. $\sqrt[3]{-27}$
3. Solve $2x + 3 < 11$
4. Simplify $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$

Warm-Up

Answers.

1. Write as one power: $3^2 \cdot 3^3$ 3^5
2. $\sqrt[3]{-27}$ -3
3. Solve $2x + 3 < 11$ $x < 4$
4. Simplify $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$ 5

Notes

Where the rule comes from.

I. Quotient of powers — same base, subtract the exponents

$$\frac{5^6}{5^3} = \frac{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}{5 \cdot 5 \cdot 5} = 5 \cdot 5 \cdot 5 = 5^3$$

Quotient of powers: $\frac{a^m}{a^n} = a^{m-n} \quad (a \neq 0)$

Check with values: $\frac{5^6}{5^3} = \frac{15625}{125} = 125 = 5^3$

Notes

II. Power of a quotient — the exponent goes top and bottom

$$\left(\frac{8}{10}\right)^3 = \frac{8}{10} \cdot \frac{8}{10} \cdot \frac{8}{10} = \frac{8 \cdot 8 \cdot 8}{10 \cdot 10 \cdot 10} = \frac{8^3}{10^3}$$

Power of a quotient: $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \quad (b \neq 0)$

Either way: $\frac{8^3}{10^3} = \frac{512}{1000} = 0.512$ and $(0.8)^3 = 0.512$

Notes

III. The zero exponent — two roads to the same answer

2^4	2^3	2^2	2^1	2^0
16	8	4	2	1

$$\frac{2^3}{2^3} = \frac{8}{8} = 1 \quad \text{and} \quad \frac{2^3}{2^3} = 2^{3-3} = 2^0 \quad \text{so} \quad 2^0 = 1$$

Zero exponent: $a^0 = 1$ for every nonzero a

Example 1

A negative base.

$$\frac{(-2)^6}{(-2)^3}$$

Which law? Then find the value.

Example 1

Worked.

$$\frac{(-2)^6}{(-2)^3} = (-2)^{6-3} = (-2)^3 \quad \text{quotient of powers: subtract}$$

$$(-2)^3 = -8 \quad \text{three negative factors}$$

$$64 \div (-8) = -8 \quad \text{check: evaluate each, then divide}$$

Answer: -8

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{3^6}{3^2}$$

Your Turn

Answer.

$$\frac{3^6}{3^2}$$

same base, subtract: 3^4

Answer: $3^4 = 81$

Example 2

A decimal base inside a quotient.

$$\left(\frac{0.2}{10}\right)^4$$

Find the value.

Example 2

Worked both ways.

$$\left(\frac{0.2}{10}\right)^4 = \frac{0.2^4}{10^4} = \frac{0.0016}{10000} \quad \text{power of a quotient}$$

$$\frac{0.0016}{10000} = 0.00000016 \quad \text{divide by } 10^4: \text{four places left}$$

$$(0.02)^4 = 0.00000016 \quad \text{check: divide first, then raise}$$

Answer: 0.00000016

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(\frac{3}{4}\right)^3$$

Your Turn

Answer.

$$\left(\frac{3}{4}\right)^3$$

numerator and denominator both cubed

Answer: $\frac{27}{64}$

Whiteboards • Question 1 of 9

Boards up on three.

$$\frac{10^5}{10^3}$$

Answer it.

Write it as one power.

Whiteboards • Question 1 of 9

Answer.

$$\frac{10^5}{10^3}$$

same base, subtract the exponents

Answer: 10^2

Whiteboards • Question 2 of 9

Boards up on three.

$$\frac{8^4}{8^1}$$

Answer it.

Find the value.

Whiteboards • Question 2 of 9

Answer.

$$\frac{8^4}{8^1}$$
$$8^3$$

Answer: 512

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{(-6)^6}{(-6)^5}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{(-6)^6}{(-6)^5}$$
$$(-6)^1$$

Answer: -6

Whiteboards • Question 4 of 9

Boards up on three.

$$\left(\frac{-5}{10}\right)^7$$

Answer it.

Apply the power of a quotient law. Do not simplify.

Whiteboards • Question 4 of 9

Answer.

$$\left(\frac{-5}{10}\right)^7$$

the exponent goes to both

Answer: $\frac{(-5)^7}{10^7}$

Whiteboards • Question 5 of 9

Boards up on three.

$$9^0$$

Which is the value?

A. 1

B. 0

C. 9

D. $\frac{1}{9}$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$9^0$$

Which is the value?

A. 1

B. 0

C. 9

D. 1/9

Answer: A — 1

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{4^8}{4^2}$$

Which expression is equivalent?

A. 4^6

B. 4^4

C. 1^6

D. 4^{10}

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\frac{4^8}{4^2}$$

Which expression is equivalent?

A. 4^6

B. 4^4

C. 1^6

D. 4^{10}

Answer: A — 4^6

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{0.5^8}{0.5^5}$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$\frac{0.5^8}{0.5^5}$$
$$0.5^3$$

Answer: 0.125

Whiteboards • Question 8 of 9

Boards up on three.

$$\left(\frac{10}{11}\right)^0$$

Answer it.

Find the value.

Whiteboards • Question 8 of 9

Answer.

$$\left(\frac{10}{11}\right)^0$$

any nonzero base to the zero power

Answer: 1

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Hannah cancels: $\frac{2^3}{2^3} = \frac{8}{8} = 1$. Sebastian subtracts: $\frac{2^3}{2^3} = 2^0$.

Who is right?

Write your answer in sentences.

Say what each one did, and what it tells you about 2^0 .

Whiteboards • Question 9 of 9

Answer.

Hannah cancels: $\frac{2^3}{2^3} = \frac{8}{8} = 1$. Sebastian subtracts: $\frac{2^3}{2^3} = 2^0$.

Who is right?

both are right, so 2^0 must equal 1

Answer: Both — $2^0 = 1$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Stop at a SmartScore of 67 on each skill. Whatever you do not finish is tonight's practice.

Today's skills

Powers with negative bases — WFW

Optional: Division with exponents — XNC

Optional: Identify equivalent expressions involving whole-number exponents — KBE